

lecture 02

na

maximum likelihood; invariance principle; predictive distribution

review

an example exam question

Simplify the following: $\log \left(\prod_{i=1}^n \pi^{y_i} (1 - \pi)^{(1-y_i)} \right)$

From the homework:

Let $\ell(\pi) = S \log(\pi) + (n - S) \log(1 - \pi)$ for $0 < \pi < 1$, where S and n (with $0 \leq S \leq n$) are fixed numerical constants. Find $\frac{d\ell(\pi)}{d\pi}$ and $\frac{d^2\ell(\pi)}{d\pi^2}$.

A variant on the exam:

Let $\ell(\pi) = \log [\pi^k (1 - \pi)^{N-k}]$ for $0 < \pi < 1$ and $0 < k < N$. Simplify the right-hand side and find $\frac{d\ell(\pi)}{d\pi}$.

I would say this question is “medium-easy” (after all, it’s review material). But it’s somewhat tedious—I’d give you 5-10 minutes to work through it.

maximum likelihood

example

the toothpaste cap problem



[illegible]

150 tosses

8 tops

How can we estimate the chance of a top?

Bernoulli

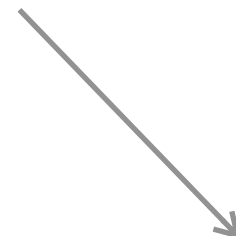
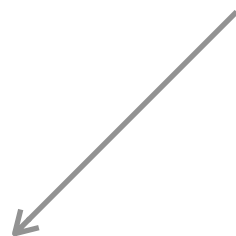
either

1

or

0

"x" for theory
"y" for applied



$$\Pr(y = 1) = \pi$$

$$\Pr(y = 0) = 1 - \pi$$

A.1 Bernoulli Distribution

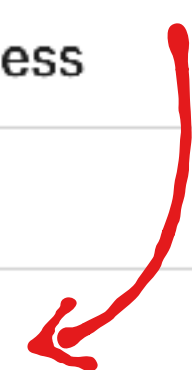
Table A.1: Summary of the Bernoulli distribution

Property	Description
Name	Bernoulli distribution
Notation	$\text{Bernoulli}(\pi)$
Parameters	$0 \leq \pi \leq 1$: probability of success
Support	$x \in \{0, 1\}$
PMF	$f(x) = \pi^x(1 - \pi)^{1-x}$
CDF	$F(x) = 0$ for $x < 0$; $1 - \pi$ for $0 \leq x < 1$; 1 for $x \geq 1$
Mean	π
Variance	$\pi(1 - \pi)$
Special cases	Binomial($n = 1, \pi$) is Bernoulli(π).

let's try this out!

$f(1) = ?$

$f(0) = ?$



Assume the observed data (i.e., those 0s and 1s) are draws from a Bernoulli distribution.

What value of π would be
most likely
to generate those data?

This is pretty easy to work out.

Remember, we have

0 0 1 0 0...

$$\begin{array}{cccccccccccc} \Pr(y_1 = 0) & \times & \Pr(y_2 = 0) & \times & \Pr(y_3 = 1) & \times & \Pr(y_4 = 0) & \times & \Pr(y_5 = 0) & \times & \dots \\ f(0) & \times & f(0) & \times & f(1) & \times & f(0) & \times & f(0) & \times & \dots \\ (1 - \pi) & \times & (1 - \pi) & \times & \pi & \times & (1 - \pi) & \times & (1 - \pi) & \times & \dots \end{array}$$

The number of π s is $k = \sum_{i=1}^N y_i$. The number of $(1 - \pi)$ is $k = N - k$.

the likelihood: $L(\pi) = \pi^k (1 - \pi)^{(N-k)}$



There's a more general way.

Remember, we have
 $y_1 \ y_2 \ y_3 \ y_4 \ y_5 \dots$

$f(x)$ can be
pmf or pdf!



$$f(y_1) \times f(y_2) \times f(y_3) \times f(y_4) \times f(y_5) \times \dots$$

the likelihood: $L(\pi) = \prod_{i=1}^N f(y_i)$

$$= \prod_{i=1}^N \pi^{y_i} (1 - \pi)^{1-y_i}.$$

= (could simplify as before if we wanted...)

the likelihood: $L(\pi) = \pi^k (1 - \pi)^{(N-k)}$

Maximize the likelihood w.r.t. π .

This is the π that would “most likely” generate our data.

MLE Recipe

1. Take log of the likelihood.
2. Simplify the log-likelihood.
3. Take derivative of the log-likelihood
4. Set derivative equal to zero; set $\pi = \hat{\pi}$.
5. Solve for $\hat{\pi}$.
6. (Check second-order conditions.)

Solution

Likelihood.

$$L(\pi) = \pi^k (1 - \pi)^{N-k}.$$

1. **Take logs.** (Use $\log(ab) = \log a + \log b$ and $\log(x^c) = c \log x$.)

$$\begin{aligned}\log L(\pi) &= \log(\pi^k (1 - \pi)^{N-k}) \\ &= \log(\pi^k) + \log((1 - \pi)^{N-k}) \\ &= k \log \pi + (N - k) \log(1 - \pi).\end{aligned}$$

2. **Differentiate $\log L(\pi)$ with respect to π .**

$$\begin{aligned}\frac{d \log L(\pi)}{d\pi} &= k \frac{d \log \pi}{d\pi} + (N - k) \frac{d \log(1 - \pi)}{d\pi} \\ &= k \left(\frac{1}{\pi} \right) + (N - k) \left(\frac{1}{1 - \pi} \frac{d(1 - \pi)}{d\pi} \right) \\ &= \frac{k}{\pi} - \frac{N - k}{1 - \pi}.\end{aligned}$$

3. **Set the derivative to zero at $\pi = \hat{\pi}$ and solve.**

$$\begin{aligned}\left. \frac{d \log L(\pi)}{d\pi} \right|_{\pi=\hat{\pi}} &= \frac{k}{\hat{\pi}} - \frac{N - k}{1 - \hat{\pi}} = 0. \\ \frac{k}{\hat{\pi}} &= \frac{N - k}{1 - \hat{\pi}} \iff k(1 - \hat{\pi}) = (N - k)\hat{\pi} \\ \iff k &= N\hat{\pi} \implies \hat{\pi} = \frac{k}{N} = \frac{1}{N} \sum_{i=1}^N y_i.\end{aligned}$$

4. **Second-order condition (concavity).** Need $\frac{d^2 \log L(\pi)}{d\pi^2} < 0$.

$$\frac{d^2 \log L(\pi)}{d\pi^2} = -\frac{k}{\pi^2} - \frac{N - k}{(1 - \pi)^2} < 0 \quad (0 < \pi < 1).$$

Thus $\log L(\pi)$ is strictly concave, and the critical point $\hat{\pi} = k/N$ is the unique global maximizer.



principle

MIL

in general

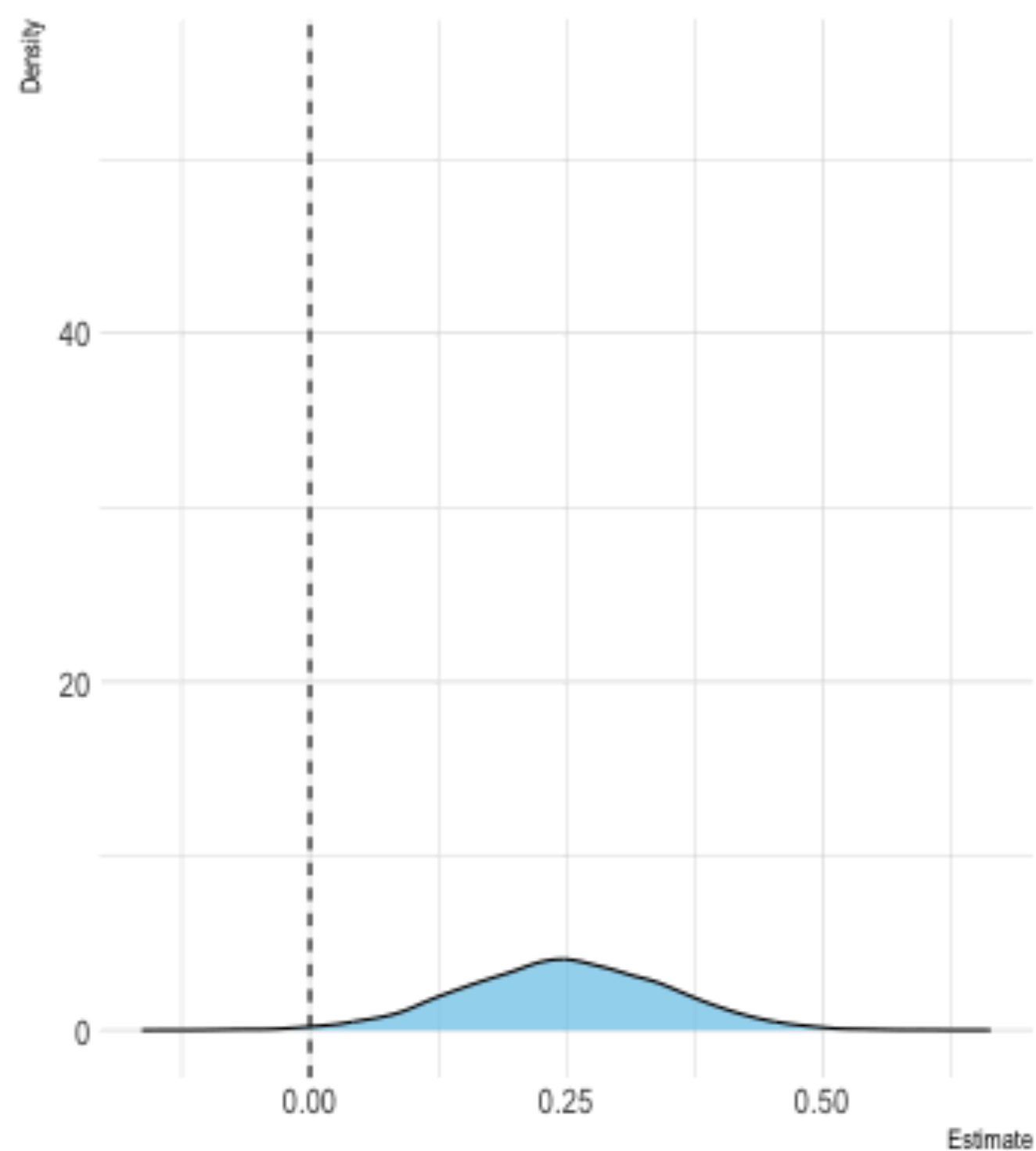
Maximum Likelihood Estimator. Suppose we have iid samples $x_1, x_2, \dots, x_N$ from pdf or pmf $f(x; \theta)$. Then the joint density/probability is $f(x; \theta) = \prod_{i=1}^N f(x_i; \theta)$ and $\log L(\theta) = \sum_{i=1}^N \log [f(x_i; \theta)]$. The ML estimator $\hat{\theta}$ of θ is $\arg \max \log L(\theta)$.

Consistency. Let $\hat{\theta}_N$ be an estimator of θ based on a sample of size N . Say that $\hat{\theta}_N$ is a **consistent** estimator for θ if $\lim_{N \rightarrow \infty} \Pr \left(|\hat{\theta}_N - \theta| \geq \varepsilon \right) = 0$ for every $\varepsilon > 0$.

Consistency of ML Estimators. Suppose an ML estimator $\hat{\theta}$ of θ . Under certain *regularity conditions*, $\hat{\theta}$ is a consistent estimator of θ .

Sampling Distribution of Estimate

Sample size: 100



example

Poisson Model

Suppose we collect N random samples $y = \{y_1, y_2, \dots, y_N\}$ and model each draw as a random variable $X \sim \text{Poisson}(\lambda)$. Find the ML estimator of λ .

Examples

Number of times legislators
travel to their districts

Number of ISIS violent events

Number of out-of-circuit
judicial citations

A.4 Poisson Distribution

Table A.4: Summary of the Poisson distribution

Property	Description
Name	Poisson distribution
Notation	$\text{Poisson}(\lambda)$
Parameters	$\lambda > 0$: rate (mean number of events per unit interval)
Support	$x \in \{0, 1, 2, \dots\}$
PMF	$f(x) = \frac{e^{-\lambda} \lambda^x}{x!}$
CDF	$F(x) = \sum_{k=0}^{\lfloor x \rfloor} \frac{e^{-\lambda} \lambda^k}{k!}$
Mean	λ
Variance	λ
Special cases	- Limit of $\text{Binomial}(n, p)$ as $n \rightarrow \infty, p \rightarrow 0$ with $np = \lambda$ fixed - Distribution of counts in a homogeneous Poisson process - Sum of independent $\text{Poisson}(\lambda_i)$ is $\text{Poisson}(\sum \lambda_i)$

Likelihood.

$$L(\lambda) = \prod_{i=1}^N \frac{e^{-\lambda} \lambda^{y_i}}{y_i!} = e^{-N\lambda} \lambda^{\sum_{i=1}^N y_i} \prod_{i=1}^N \frac{1}{y_i!}.$$

1. **Take logs.** (Use $\log(\prod a_i) = \sum \log a_i$, $\log(ab) = \log a + \log b$, and $\log(x^c) = c \log x$.)

$$\begin{aligned} \log L(\lambda) &= \sum_{i=1}^N [-\lambda + y_i \log \lambda - \log(y_i!)] \\ &= -N\lambda + \left(\sum_{i=1}^N y_i \right) \log \lambda - \sum_{i=1}^N \log(y_i!). \end{aligned}$$

2. **Differentiate $\log L(\lambda)$ with respect to λ .**

$$\begin{aligned} \frac{d \log L(\lambda)}{d\lambda} &= \frac{d}{d\lambda} [-N\lambda] + \frac{d}{d\lambda} \left[\left(\sum_{i=1}^N y_i \right) \log \lambda \right] - \frac{d}{d\lambda} \left[\sum_{i=1}^N \log(y_i!) \right] \\ &= -N + \left(\sum_{i=1}^N y_i \right) \frac{1}{\lambda} + 0 \\ &= -N + \frac{\sum_{i=1}^N y_i}{\lambda}. \end{aligned}$$

3. **Set the derivative to zero at $\lambda = \hat{\lambda}$ and solve.**

$$\begin{aligned} \left. \frac{d \log L(\lambda)}{d\lambda} \right|_{\lambda=\hat{\lambda}} &= -N + \frac{\sum_{i=1}^N y_i}{\hat{\lambda}} = 0. \\ \frac{\sum_{i=1}^N y_i}{\hat{\lambda}} = N &\iff \sum_{i=1}^N y_i = N\hat{\lambda} \implies \hat{\lambda} = \frac{1}{N} \sum_{i=1}^N y_i = \bar{y}. \end{aligned}$$

4. **Second-order condition (concavity).** Need $\frac{d^2 \log L(\lambda)}{d\lambda^2} < 0$.

$$\frac{d^2 \log L(\lambda)}{d\lambda^2} = -\frac{\sum_{i=1}^N y_i}{\lambda^2} < 0 \quad (\lambda > 0 \text{ and } \sum_{i=1}^N y_i > 0).$$

Thus $\log L(\lambda)$ is strictly concave and the critical point $\hat{\lambda} = \bar{y}$ is the unique global maximizer.

invariance property

Invariance Property. Suppose an ML estimator $\hat{\theta}$ of θ and a quantity of interest $\tau = \tau(\theta)$ for any function τ . The ML estimate $\hat{\tau}$ of $\tau = \tau(\theta)$ is $\tau(\hat{\theta})$.

Example 1: $\hat{\pi}$ to odds

Example 2: $\hat{\lambda}$ to SD



optim()

example

Beta Model

A.2 Beta Distribution

Table A.2: Summary of the beta distribution

Property	Description
Name	beta distribution
Notation	$\text{beta}(\alpha, \beta)$
Parameters	$\alpha > 0$: shape; $\beta > 0$: shape
Support	$x \in (0, 1)$
PDF	$f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}$
CDF	$F(x) = I_x(\alpha, \beta)$, the regularized incomplete beta function
Mean	$\frac{\alpha}{\alpha + \beta}$
Variance	$\frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$
Special cases	$\text{beta}(1, 1)$ is $\text{uniform}(0, 1)$.

$$L(\alpha, \beta) = \prod_{i=1}^n \frac{y_i^{\alpha-1} (1 - y_i)^{\beta-1}}{B(\alpha, \beta)}$$

$$\log L(\alpha, \beta) = \sum_{i=1}^n \log \frac{y_i^{\alpha-1} (1 - y_i)^{\beta-1}}{B(\alpha, \beta)}$$

$$= \sum_{i=1}^n [\log y_i^{\alpha-1} + \log(1 - y_i)^{\beta-1} - \log B(\alpha, \beta)]$$

$$= \sum_{i=1}^n [(\alpha - 1) \log y_i + (\beta - 1) \log(1 - y_i) - \log B(\alpha, \beta)]$$

$$= \sum_{i=1}^n [(\alpha - 1) \log y_i + (\beta - 1) \log(1 - y_i)] - n \log B(\alpha, \beta)$$

$$\log L(\alpha, \beta) = (\alpha - 1) \sum_{i=1}^n \log y_i + (\beta - 1) \sum_{i=1}^n \log(1 - y_i) - n \log B(\alpha, \beta)$$

$$\log L(\alpha, \beta) = (\alpha - 1) \sum_{i=1}^n \log y_i + (\beta - 1) \sum_{i=1}^n \log(1 - y_i) - n \log B(\alpha, \beta)$$

manually typing log-likelihood

```
ll_fn <- function(theta, y) {
  alpha <- theta[1]
  beta <- theta[2]
  ll <- alpha*sum(log(y)) + beta*sum(log(1 - y)) -
    length(y)*log(beta(alpha, beta))
  return(ll)
}
```

using dbeta shortcut (good!)

```
ll_fn <- function(theta, y) {
  alpha <- theta[1]
  beta <- theta[2]
  ll <- sum(dbeta(y, shape1 = alpha, shape2 = beta, log = TRUE))
  return(ll)
}
```

```
# using dbeta shortcut (good!)
ll_fn <- function(theta, y) {
  alpha <- theta[1]
  beta <- theta[2]
  ll <- sum(dbeta(y, shape1 = alpha, shape2 = beta, log = TRUE))
  return(ll)
}
```

```
# use optim()
est <- optim(par = c(2, 2),
            fn = ll_fn,
            y = y,
            control = list(fnscale = -1),
            method = "Nelder-Mead")
```

```
# using dbeta shortcut (good!)
```

```
ll_fn <- function(theta, y) {  
  alpha <- theta[1]  
  beta <- theta[2]  
  ll <- sum(dbeta(y, shape1 = alpha, shape2 = beta, log = TRUE))  
  return(ll)  
}
```

```
# make function that fits beta model
```

```
est_beta <- function(y) {  
  est <- optim(par = c(2, 2), fn = ll_fn, y = y,  
              control = list(fnscale = -1),  
              method = "BFGS") # for >1d problems  
  if (est$convergence != 0) print("Model did not converge!")  
  res <- list(est = est$par)  
  return(res)  
}
```

```
# fit beta model
```

```
ml_est <- est_beta(y)  
print(ml_est, digits = 3)
```

(Continue in R—there's a Gist.)

<https://gist.github.com/carlislerainey/17f281357ffd91972dce3dc670f35a37>

predictive distribution

In my view, the predictive distribution is the best way to (1) understand, (2) evaluate, and then (3) improve models.

You can use the predictive distribution as follows:

1. Fit your model with maximum likelihood.
2. Simulate a new outcome variable using the estimated model parameters (i.e., $f(y; \hat{\theta})$). Perhaps simulate several for comparison.
3. Compare the simulated outcome variable(s) to the observed outcome variable. You can use histograms, plots of the ECDF, or other summaries.

Example: Holland (2015)

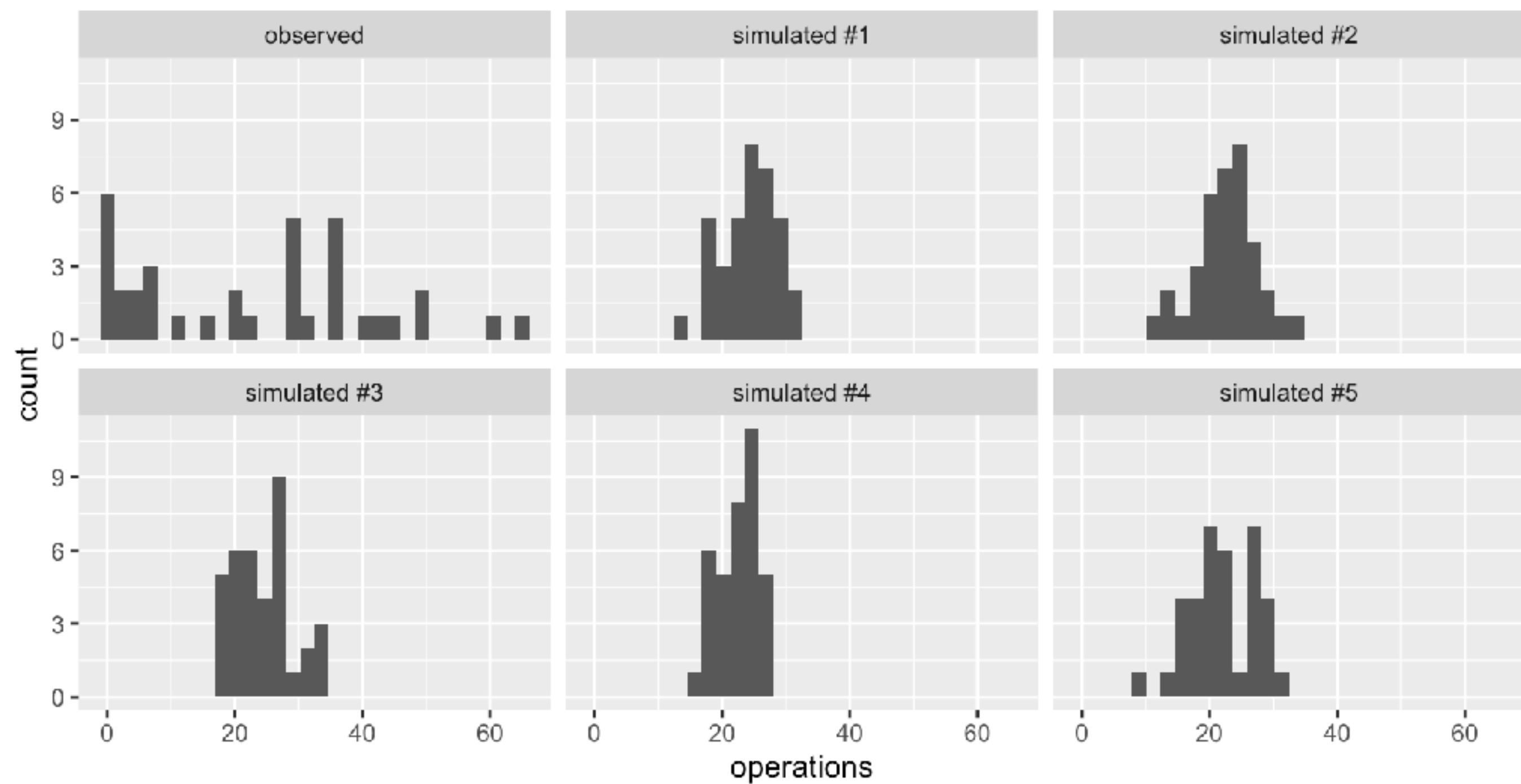
number of enforcement operations in a city

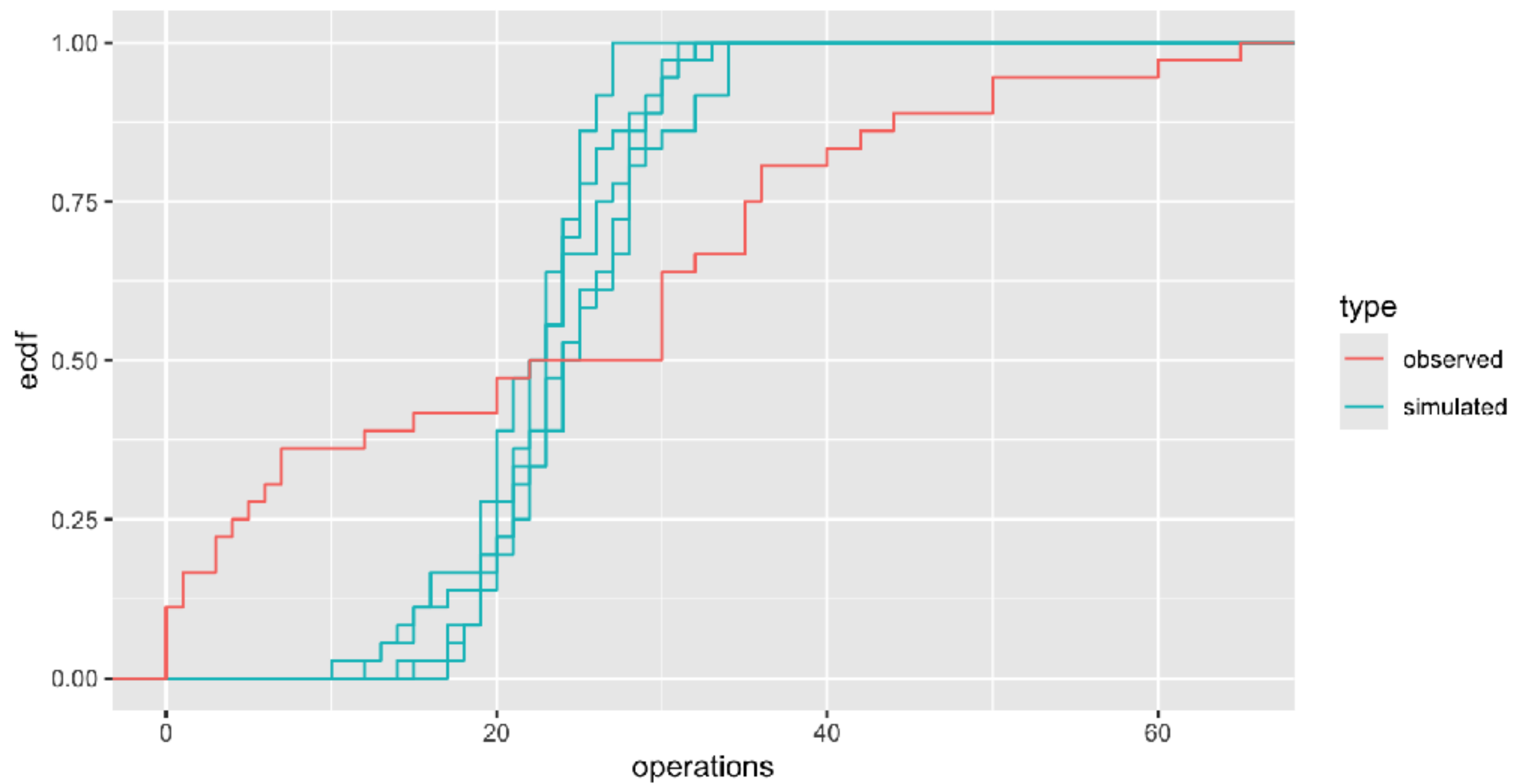
Observed data:

7 0 20 1 1 50 3 0 36 32...

Simulated data:

23 25 20 22 19 19 22 8 14 25...





(Continue in R—there's a Gist.)

<https://gist.github.com/carlislerainey/c3cb055454a076ab49ebca1eca6b0c0c>

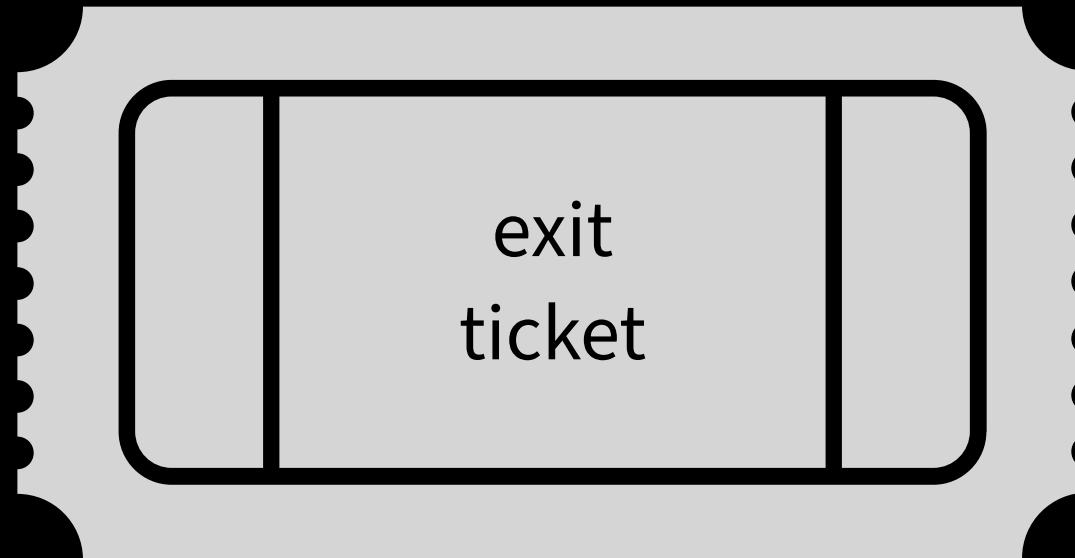


a bit of fun

German Tank Problem

Suppose a discrete uniform distribution from 0 to K . The pdf is $f(x; K) = \frac{1}{K}$ for $x \in \{0, 1, \dots, K\}$. Suppose I have three samples from the distribution: 276, 159, and 912.

1. Find the ML estimate of K . *Hint: The log-likelihood is discontinuous, so the usually optimization routine might mislead you. But the maximum is immediately apparent once you write out the likelihood.*
2. Find the method of moments estimate of K . *Hint: The mean of this uniform distribution is $\frac{K}{2}$.*
3. Discuss any problems you notice with each estimator.



List three important ideas from today's class. For each, briefly connect it to something we studied in an earlier class.